

KCPC Week 3 Reading

The Lambda Calculus: Functions, Computation, and Numbers

Follow this guide alongside the slides and problem sheet. Start with the supplied Oxford notes, then use the accessible introductions for extra worked examples; the research bibliography in *Week 3 Papers* gives the original sources. Page numbers below refer to the *printed* pages of each text, rather than the PDF viewer counter.

I. The supplied further reading: Oxford lecture notes

Lambda Calculus and Types (Andrew D. Ker)

Oxford University Computing Laboratory, 16 lectures, Hilary Term 2009. Available locally at the supplied Oxford lecture-notes PDF (in [reading/](#)).

- **Chapter 1, §§1.2–1.5, pp. 2–7:** the grammar of terms, application association, free/bound variables, legal alpha-conversion and capture-avoiding substitution. Do this before Problems 1–5.
- **Chapter 4, §4.1, pp. 43–44:** introduces Boolean selectors, Church numerals, successor and the zero test, and states the required predecessor properties. Compare with Problems 6–10. The chapter's introduction explains why numbers are encoded instead of built in.
- **Exercises 4.1–4.2, p. 50:** construct AND and verify a proposed predecessor, then repair its zero case. These are proof exercises, not worked solutions; compare them with Problems 6 and 9 and use Selinger for the arithmetic definitions.
- **Chapter 2, §§2.1–2.5, pp. 15–23 (after the workshop):** reduction sequences and the Church–Rosser theorem; why choosing different beta redexes need not change a normal form.
- **Chapter 3, §3.1, pp. 31–33 (extension):** reduction strategies and why a concrete evaluator must choose which redex to contract.

The appendix on computational practice (pp. 109–115) gives answers to the notes' short calculation exercises; use it to check your attempts rather than to replace a proof.

II. Accessible introductions and worked examples

Lecture Notes on the Lambda Calculus (Peter Selinger)

Dalhousie University, revised 2013, arXiv:0804.3434. <https://arxiv.org/abs/0804.3434>

- **§2:** untyped terms, binding, substitution, alpha- and beta-conversion (Problems 1–5).
- **§§3.1–3.2:** Boolean selectors, Church numerals, successor and arithmetic, with reduction arguments and exercises (Problems 6–10).
- **§3.4:** encodings of pairs and other data types; use the pair projections alongside Problem 9.
- **§4:** further theory of reduction; optional once the worksheet is complete.

Types and Programming Languages (Benjamin C. Pierce)

MIT Press, 2002. Contents and chapter locations: <https://www.cis.upenn.edu/~bcpierce/tapl/contents.pdf>

- **Chapter 5, §5.1, pp. 51–58:** untyped lambda syntax, substitution and evaluation; clarifies exactly what a binder does.
- **Chapter 5, §5.2, pp. 58–68:** currying, Booleans, pairs, Church numerals and recursion; alternate encoding examples.
- **Chapters 6–7, pp. 75–88 (optional):** nameless variables and an implementation of the calculus, particularly helpful if you want to build an evaluator after the two extended C++ homework tasks.

Church Numerals and Booleans (OpenDSA)

Virginia Tech, Programming Languages, §3.8. <https://opensa-server.cs.vt.edu/ODSA/Books/PL/html/ChurchNumerals.html> Interactive reduction traces for AND, successor and multiplication, plus definitions of T/F, predecessor and ISZERO. Use the animations to check a small input, then use iteration and induction to justify your answer for *every* numeral.

III. Suggested order

Read Ker §§1.2–1.5, solve Problems 1–5; read Ker §4.1, Selinger §§3.1–3.2 and OpenDSA §3.8, solve Problems 6–10. For the pair construction in Problem 9, consult Selinger §3.4 or Pierce §5.2 and compare Ker Exercise 4.2. Then prove the stack invariants in Problems 11–12. For original research and precise bibliographic locations see *Week 3 Papers*.

